# Analysis

## Time Complexity

The method makes one recursive call for each year.

- **Time Complexity:** O(n)
- **Space Complexity:** O(n) due to the recursion stack.

where n is the number of forecast years.

### How to Optimize the Recursive Solution

```
public static double predictFutureValue(double currentAmount,
                                        double annualGrowthRate,
                                        int forecastYears) {

    for (int year = 1; year <= forecastYears; year++) {
        currentAmount *= (1 + annualGrowthRate);
    }

    return currentAmount;
}
```

Recursion simplifies financial forecasting by expressing future value as repeated growth over smaller time periods. The recursive solution has O(n) time complexity and O(n) space complexity. For very large forecasting periods, an iterative approach is preferred because it achieves the same O(n) running time while reducing space complexity to O(1).